

1. Questions

Read the following information carefully and answer the questions based on it.

The table chart given below shows the average number of males and females in five cities i.e. P, Q, R, S and T. It also shows the average number of females and children in each city.

City	The average number of males and females	The average number of females and children
P	100	65
Q	125	75
R	75	45
S	195	90
T	180	80

Note:

- I). The total number of people = the number of males + the number of females + the number of children
- II). The ratio of the number of males in cities P and Q is 4:5.
- III). The number of children in cities P and Q are the same.
- IV). The number of females in city R is 60% less than in city S.
- V). The difference between the number of males in cities R and S is 150. ($R < S$)
- VI). The total number of people in city T is 400.

Out of the total number of children in cities P and S, some are boys and the rest are girls. If the ratio of the number of boys in cities P and S is 7:4 and the number of girls in city P is 5 more than in city S, then find the difference between the number of boys and girls in city S.

- a. 18
- b. 12
- c. 10
- d. 16
- e. 8

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2. Questions

In city Q, if $\frac{1}{3}$ rd of the total number of males who work in public sector, which is 25% more than the number of females who work in public sector, then find the ratio of the number of males and females who work in private sector in city Q.

- a. 3:5

- b. 5:4
- c. 4:5
- d. 5:3
- e. 3:7

3. Questions

The total number of people in another city U is 14.28% less than in city S and the number of children in city U is 60. In city U, if the average age of all males and females are 30 years and 24 years respectively, while the total age all males and females are 8400 years, then find the number of females in city U.

- a. 100
- b. 200
- c. 240
- d. 160
- e. 80

4. Questions

Out of the total number of children in cities R and T, some are in primary school, while the rest are in secondary school. If the number of children in secondary school in cities R and T are same and the ratio of the number of students in primary school in cities R and T is 9:14, then find the number of students in primary school in city R.

- a. 12
- b. 18
- c. 16
- d. Cannot be determined
- e. None of these

5. Questions

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In city T, out of the total number of males and females, $\frac{1}{4}$ th did not attend the seminar. If the number of males who attended the seminar in city T is 200, then how much percent of the number of females did not attend the seminar in city T?

- a. 41.67%
- b. 16.67%
- c. 37.50%
- d. 8.33%

e. 11.11%

6. Questions

A started a business with Rs. 1500 and after _____ months, B also joined him with _____% more than by A's investment. At the end of the year, the ratio of the profit share received by A and B is _____.

Which of the following statement(s) is/are satisfied the fill-up(s) in the same order?

I: 7, 40 and 12:7

II: 8, 20 and 5:2

III: 6, 80 and 11:8

- a. Only I follow
- b. Both I and II follow
- c. Both II and III follow
- d. All I, II and III follow
- e. None follow

7. Questions

If an article, whose cost price is Rs. _____ is sold at a profit of _____% after allowing a discount of Rs. 300, then the marked price of the same article will be Rs. _____.

Which of the following statement(s) is/are satisfied the fill-up(s) in the same order?

I: 2000, 35 and 3000

II: 1000, 80 and 2400

III: 4000, 20 and 5000

- a. Only I follow
- b. Both I and II follow
- c. Both II and III follow
- d. All I, II and III follow
- e. None follow

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8. Questions

The speed of the river is _____ km/hr, which is $\frac{1}{4}^{\text{th}}$ of the speed of the boat in still water. The time taken by the boat to cover 120 km downstream is equal to the time taken by the same boat to cover _____ km upstream.

Which of the following statement(s) is/are satisfied the fill-up(s) in the same order?

I: 6 and 60

II: 8 and 72

III: 4 and 48

- a. Only III follow
- b. Both I and II follow
- c. Only II follow
- d. All I, II and III follow
- e. None follow

9. Questions

The amount invested in scheme A is Rs. 40000, which offers simple interest at the rate of _____ % per annum for 4 years. The interest received from scheme A is further invested in scheme B, which offers compound interest at the rate of 20% per annum for 2 years. The interest received from scheme B is Rs. _____.

Which of the following statement(s) is/are satisfied the fill-up(s) in the same order?

I: 12 and 8448

II: 16 and 13200

III: 8 and 5632

- a. Only I follow
- b. Both I and III follow
- c. Only II follow
- d. All I, II and III follow
- e. None follow

10. Questions

The time taken by train P of length _____ meters to cross a platform of length 120 meters with the speed of 40 m/sec is 2 seconds more than the time taken by train Q of length _____ meters to cross the same platform with the speed of 60 m/sec.

Which of the following statement(s) is/are satisfied the fill-up(s) in the same order?

I: 240 and 300

II: 280 and 360

III: 360 and 480

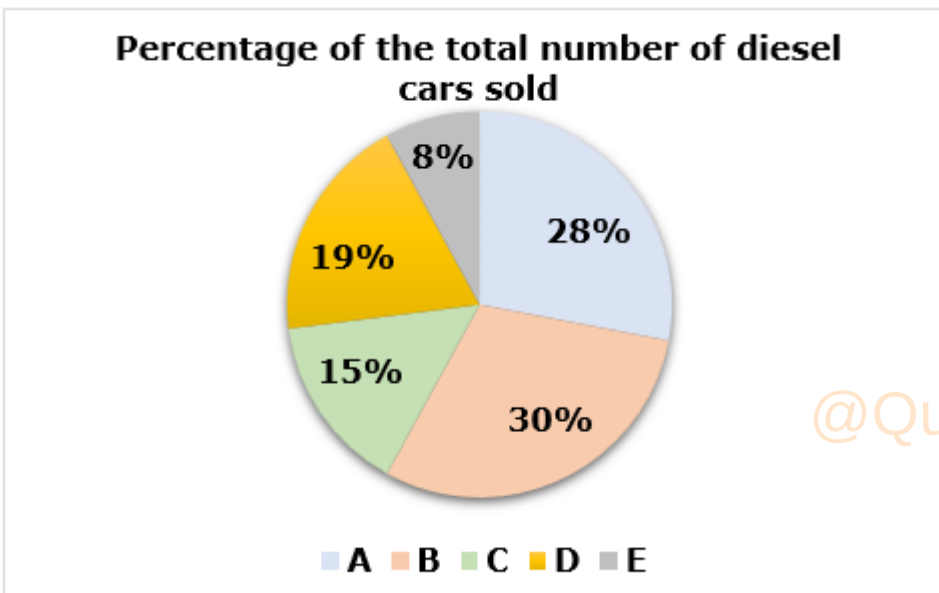
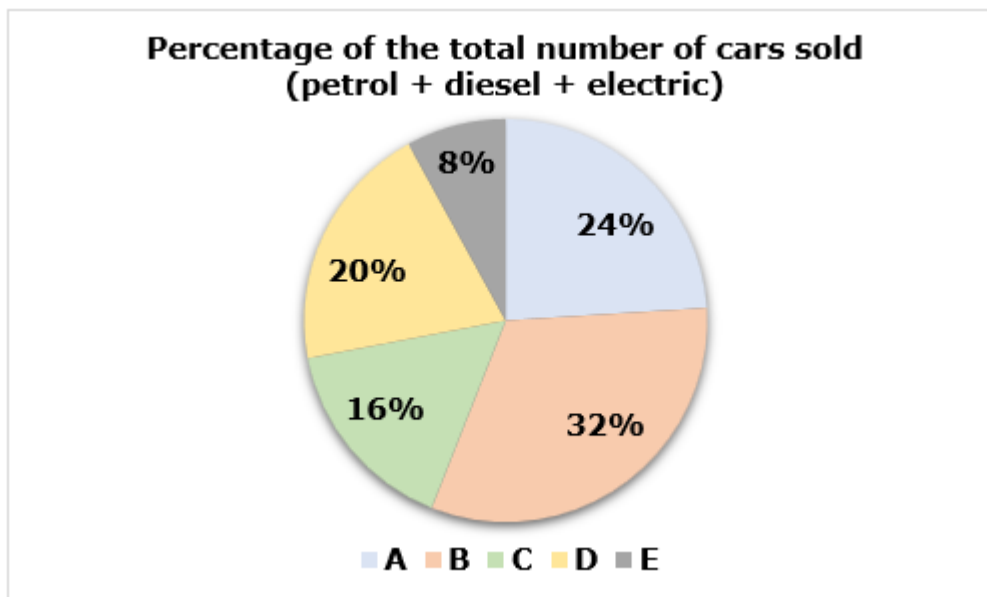
- a. Only I follow
- b. Both I and III follow

- c. Only II follow
- d. All I, II and III follow
- e. None follow

11. Questions

Read the following information carefully and answer the questions based on it.

The first pie chart given below shows the percentage distribution of the total number of cars sold (petrol + diesel + electric) by five shops A, B, C, D and E. The second pie chart given below shows the percentage distribution of the total number of diesel cars sold by each shop.



Note:

I). The total number of cars sold = the number of petrol cars sold + the number of diesel cars sold + the number of electric cars sold

II). $\frac{2}{5}$ th of the total number of cars sold by all shops together are diesel cars.

III). The number of electric cars sold by shops A, B, C, D and E is 144, 200, 60, 124 and 24 respectively.

IV). The difference between the number of petrol cars sold by shops A and B is 160. ($A < B$)

If 25% of the total number of petrol cars sold by shop E are sold to females, which is $\frac{3}{8}$ th of the number of petrol cars sold to females by shop C, then find the difference between the number of petrol cars sold to males by shops C and E.

- a. 70
- b. 64
- c. 48
- d. 50
- e. 84

12. Questions

In shop B, if the total number of petrol cars manufactured is 20 more than diesel cars and the ratio of the number of petrol and diesel cars unsold is 5:6, then find the total number of diesel cars manufactured by shop B.

- a. 560
- b. 800
- c. 640
- d. 480
- e. 720

13. Questions

The total number of cars sold by another shop F is 100 less than that by shop D and the ratio of the sum of petrol and diesel cars sold to the sum of diesel and electric cars sold by shop F is 8:5. If the number of petrol cars sold by shop F is 250, then find the number of electric cars sold by the same shop.

- a. 120
- b. 100
- c. 240
- d. 300
- e. 150

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14. Questions

Out of the total number of cars (petrol + diesel + electric) manufactured by shop E, $\frac{1}{4}$ of the number remained unsold and the number of cars sold by shop E is 80 more than the number of cars unsold

by the same shop.

What should come in place of question mark?

- a. 20
- b. 60
- c. 40
- d. 80
- e. 54

15. Questions

In shop A, if $\frac{1}{5}^{\text{th}}$ of the total number of diesel cars manufactured are defective and the number of non-defective diesel cars unsold is 144, then find the total number of diesel cars manufactured by shop A. (Only non-defective diesel cars can be sold)

- a. 480
- b. 600
- c. 720
- d. 840
- e. 660

16. Questions

Find the approximate value and determine what should come in place of the question mark?

$$(29.99\% \text{ of } 720.01) + (40.03\% \text{ of } 359.97) = ? + (49.89\% \text{ of } 499.92) + (10.08\% \text{ of } 1000.06)$$

- a. 26
- b. 10
- c. 30
- d. 18
- e. 7

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17. Questions

$$(23.95 * 8.06) \div (12.09 * 3.93) + ? = (34.98 * 4.96) - (23.04 * 7.01)$$

- a. 10
- b. 24
- c. 33
- d. 7

e. 16

18. Questions

$$(\sqrt{24.89\% \text{ of } 1200.11}) + (\sqrt{16.13\% \text{ of } 999.88}) = (12.09\% \text{ of } 499.96) + (19.91\% \text{ of } 400.09) - ?$$

- a. 52
- b. 36
- c. 40
- d. 24
- e. 48

19. Questions

$$(63.91/15.94) * (90.04/9.02) = ? + (87.93/3.98) * (45.01/9.07)$$

- a. 70
- b. 35
- c. -35
- d. -70
- e. 0

20. Questions

$$[(390.01/2.99) + (119.93/4.07)] - [(249.96/5.11) + (140.06/13.90)] = ?^2$$

- a. 12
- b. 18
- c. 16
- d. 14
- e. 10

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Explanations:

1. Questions

The total number of males and females in,

$$\text{City P} = (100 * 2) = 200$$

$$\text{City Q} = (125 * 2) = 250$$

$$\text{City R} = (75 * 2) = 150$$

$$\text{City S} = (195 * 2) = 390$$

$$\text{City T} = (180 * 2) = 360$$

The total number of females and children in,

$$\text{City P} = (65 * 2) = 130$$

$$\text{City Q} = (75 * 2) = 150$$

$$\text{City R} = (45 * 2) = 90$$

$$\text{City S} = (90 * 2) = 180$$

$$\text{City T} = (80 * 2) = 160$$

According to the note II,

Now,

Let the number of males in cities P and Q be $4k$ and $5k$ respectively.

So, the number of females in city P = $(200 - 4k)$

And, the number of children in city P = $\{130 - (200 - 4k)\} = (4k - 70)$

Also,

The number of females in city Q = $(250 - 5k)$

And, the number of children in city Q = $\{150 - (250 - 5k)\} = (5k - 100)$

Now,

$$(4k - 70) = (5k - 100)$$

$$k = 30$$

According to the note IV,

Also,

The ratio of the number of females in cities R and S = $40:100 = 2:5$. Let it be $2n$ and $5n$ respectively.

So, the number of males in city R = $(150 - 2n)$

And, the number of children in city R = $(90 - 2n)$

Also,

The number of males in city S = $(390 - 5n)$

And, the number of children in city S = $(180 - 5n)$

According to the note V,

Now,

$$(390 - 5n) - (150 - 2n) = 150$$

$$240 - 3n = 150$$

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$$3n = 90$$

$$n = 30$$

City P

$$\text{The number of males} = 4k = (4 * 30) = 120$$

$$\text{The number of females} = (200 - 120) = 80$$

$$\text{The number of children} = (120 - 70) = 50$$

$$\text{The total number of people} = (120 + 80 + 50) = 250$$

City Q

$$\text{The number of males} = 5k = (5 * 30) = 150$$

$$\text{The number of females} = (250 - 150) = 100$$

$$\text{The number of children} = (150 - 100) = 50$$

$$\text{The total number of people} = (150 + 100 + 50) = 300$$

City R

$$\text{The number of females} = 2n = (2 * 30) = 60$$

$$\text{The number of males} = (150 - 60) = 90$$

$$\text{The number of children} = (90 - 60) = 30$$

$$\text{The total number of people} = (60 + 90 + 30) = 180$$

City S

$$\text{The number of females} = 5n = (5 * 30) = 150$$

$$\text{The number of males} = (390 - 150) = 240$$

$$\text{The number of children} = (180 - 150) = 30$$

$$\text{The total number of people} = (150 + 240 + 30) = 420$$

City T

Let the number of females be m .

$$\text{So, the number of males} = (360 - m)$$

$$\text{And, the number of children} = (160 - m)$$

According to the note VI,

$$\text{The total number of people} = 400$$

Now,

$$m + (360 - m) + (160 - m) = 400$$

$$520 - m = 400$$

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$$m = 120$$

Therefore,

$$\text{The number of females} = m = 120$$

$$\text{The number of males} = (360 - m) = (360 - 120) = 240$$

$$\text{The number of children} = (160 - m) = (160 - 120) = 40$$

City	The total number of people	The number of males	The number of females	The number of children
P	250	120	80	50
Q	300	150	100	50
R	180	90	60	30
S	420	240	150	30
T	400	240	120	40
Total	1550	840	510	200

Answer: C

Now, according to the question,

$$\text{The total number of children in city P} = 50$$

$$\text{The total number of children in city S} = 30$$

Let the number of boys in city P and S be $7a$ and $4a$ respectively.

$$\text{So, the number of girls in city P} = (50 - 7a)$$

$$\text{And, the number of girls in city S} = (30 - 4a)$$

Now,

$$(50 - 7a) - (30 - 4a) = 5$$

$$20 - 3a = 5$$

$$3a = 15$$

$$a = 5$$

$$\text{So, the number of boys in city S} = 4a = (4 * 5) = 20$$

$$\text{And, the number of girls in city S} = (30 - 4a) = (30 - 20) = 10$$

$$\text{Therefore, the required difference} = (20 - 10) = 10.$$

Hence, the correct answer is option C.

2. Questions

The total number of males and females in,

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$$\text{City P} = (100 * 2) = 200$$

$$\text{City Q} = (125 * 2) = 250$$

$$\text{City R} = (75 * 2) = 150$$

$$\text{City S} = (195 * 2) = 390$$

$$\text{City T} = (180 * 2) = 360$$

The total number of females and children in,

$$\text{City P} = (65 * 2) = 130$$

$$\text{City Q} = (75 * 2) = 150$$

$$\text{City R} = (45 * 2) = 90$$

$$\text{City S} = (90 * 2) = 180$$

$$\text{City T} = (80 * 2) = 160$$

According to the note II,

Now,

Let the number of males in cities P and Q be $4k$ and $5k$ respectively.

So, the number of females in city P = $(200 - 4k)$

And, the number of children in city P = $\{130 - (200 - 4k)\} = (4k - 70)$

Also,

The number of females in city Q = $(250 - 5k)$

And, the number of children in city Q = $\{150 - (250 - 5k)\} = (5k - 100)$

Now,

$$(4k - 70) = (5k - 100)$$

$$k = 30$$

According to the note IV,

Also,

The ratio of the number of females in cities R and S = $40:100 = 2:5$. Let it be $2n$ and $5n$ respectively.

So, the number of males in city R = $(150 - 2n)$

And, the number of children in city R = $(90 - 2n)$

Also,

The number of males in city S = $(390 - 5n)$

And, the number of children in city S = $(180 - 5n)$

According to the note V,

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Now,

$$(390 - 5n) - (150 - 2n) = 150$$

$$240 - 3n = 150$$

$$3n = 90$$

$$n = 30$$

City P

$$\text{The number of males} = 4k = (4 * 30) = 120$$

$$\text{The number of females} = (200 - 120) = 80$$

$$\text{The number of children} = (120 - 70) = 50$$

$$\text{The total number of people} = (120 + 80 + 50) = 250$$

City Q

$$\text{The number of males} = 5k = (5 * 30) = 150$$

$$\text{The number of females} = (250 - 150) = 100$$

$$\text{The number of children} = (150 - 100) = 50$$

$$\text{The total number of people} = (150 + 100 + 50) = 300$$

City R

$$\text{The number of females} = 2n = (2 * 30) = 60$$

$$\text{The number of males} = (150 - 60) = 90$$

$$\text{The number of children} = (90 - 60) = 30$$

$$\text{The total number of people} = (60 + 90 + 30) = 180$$

City S

$$\text{The number of females} = 5n = (5 * 30) = 150$$

$$\text{The number of males} = (390 - 150) = 240$$

$$\text{The number of children} = (180 - 150) = 30$$

$$\text{The total number of people} = (150 + 240 + 30) = 420$$

City T

Let the number of females be m .

$$\text{So, the number of males} = (360 - m)$$

$$\text{And, the number of children} = (160 - m)$$

According to the note VI,

$$\text{The total number of people} = 400$$

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Now,

$$m + (360 - m) + (160 - m) = 400$$

$$520 - m = 400$$

$$m = 120$$

Therefore,

$$\text{The number of females} = m = 120$$

$$\text{The number of males} = (360 - m) = (360 - 120) = 240$$

$$\text{The number of children} = (160 - m) = (160 - 120) = 40$$

City	The total number of people	The number of males	The number of females	The number of children
P	250	120	80	50
Q	300	150	100	50
R	180	90	60	30
S	420	240	150	30
T	400	240	120	40
Total	1550	840	510	200

Answer: D

In city Q,

$$\text{The total number of males} = 150$$

$$\text{The number of males who work in public sector} = (1/3) * 150 = 50$$

$$\text{The number of males who work in private sector} = (2/3) * 150 = 100$$

Also,

$$\text{The total number of females} = 100$$

$$\text{The number of females who work in public sector} = 50 * (100/125) = 40$$

$$\text{The number of males who work in private sector} = (100 - 40) = 60$$

$$\text{Therefore, the required ratio} = 100:60 = 5:3.$$

Hence, the correct answer is option D.

3. Questions

The total number of males and females in,

$$\text{City P} = (100 * 2) = 200$$

$$\text{City Q} = (125 * 2) = 250$$

$$\text{City R} = (75 * 2) = 150$$

$$\text{City S} = (195 * 2) = 390$$

$$\text{City T} = (180 * 2) = 360$$

The total number of females and children in,

$$\text{City P} = (65 * 2) = 130$$

$$\text{City Q} = (75 * 2) = 150$$

$$\text{City R} = (45 * 2) = 90$$

$$\text{City S} = (90 * 2) = 180$$

$$\text{City T} = (80 * 2) = 160$$

According to the note II,

Now,

Let the number of males in cities P and Q be $4k$ and $5k$ respectively.

So, the number of females in city P = $(200 - 4k)$

And, the number of children in city P = $\{130 - (200 - 4k)\} = (4k - 70)$

Also,

The number of females in city Q = $(250 - 5k)$

And, the number of children in city Q = $\{150 - (250 - 5k)\} = (5k - 100)$

Now,

$$(4k - 70) = (5k - 100)$$

$$k = 30$$

According to the note IV,

Also,

The ratio of the number of females in cities R and S = $40:100 = 2:5$. Let it be $2n$ and $5n$ respectively.

So, the number of males in city R = $(150 - 2n)$

And, the number of children in city R = $(90 - 2n)$

Also,

The number of males in city S = $(390 - 5n)$

And, the number of children in city S = $(180 - 5n)$

According to the note V,

Now,

$$(390 - 5n) - (150 - 2n) = 150$$

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$$240 - 3n = 150$$

$$3n = 90$$

$$n = 30$$

City P

$$\text{The number of males} = 4k = (4 * 30) = 120$$

$$\text{The number of females} = (200 - 120) = 80$$

$$\text{The number of children} = (120 - 70) = 50$$

$$\text{The total number of people} = (120 + 80 + 50) = 250$$

City Q

$$\text{The number of males} = 5k = (5 * 30) = 150$$

$$\text{The number of females} = (250 - 150) = 100$$

$$\text{The number of children} = (150 - 100) = 50$$

$$\text{The total number of people} = (150 + 100 + 50) = 300$$

City R

$$\text{The number of females} = 2n = (2 * 30) = 60$$

$$\text{The number of males} = (150 - 60) = 90$$

$$\text{The number of children} = (90 - 60) = 30$$

$$\text{The total number of people} = (60 + 90 + 30) = 180$$

City S

$$\text{The number of females} = 5n = (5 * 30) = 150$$

$$\text{The number of males} = (390 - 150) = 240$$

$$\text{The number of children} = (180 - 150) = 30$$

$$\text{The total number of people} = (150 + 240 + 30) = 420$$

City T

Let the number of females be m .

$$\text{So, the number of males} = (360 - m)$$

$$\text{And, the number of children} = (160 - m)$$

According to the note VI,

$$\text{The total number of people} = 400$$

Now,

$$m + (360 - m) + (160 - m) = 400$$

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$$520 - m = 400$$

$$m = 120$$

Therefore,

$$\text{The number of females} = m = 120$$

$$\text{The number of males} = (360 - m) = (360 - 120) = 240$$

$$\text{The number of children} = (160 - m) = (160 - 120) = 40$$

City	The total number of people	The number of males	The number of females	The number of children
P	250	120	80	50
Q	300	150	100	50
R	180	90	60	30
S	420	240	150	30
T	400	240	120	40
Total	1550	840	510	200

Answer: A

In city U

$$\text{The total number of people} = [1 - (1/7)] * 420 \text{ (Note: } 14.28\% = 1/7)$$

$$= (6/7) * 420 = 360$$

According to the question,

$$\text{The number of children} = 60$$

$$\text{So, the sum of the number of males and females} = 360 - 60 = 300.$$

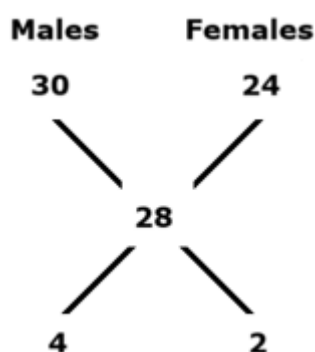
$$\text{The average age of all males} = 30 \text{ years}$$

$$\text{The average age of all females} = 24 \text{ years}$$

$$\text{So, the average all males and females} = (8400/300) = 28 \text{ years}$$

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Now, using the allegation method, we get,



So, the ratio of the number of males and females in city U = $4:2 = 2:1$.

Therefore, the number of males in city U = $(1/3) * 300 = 100$.

Hence, the correct answer is option A.

Another solution:

In city U

The total number of people = $[1 - (1/7)] * 420$ (**Note: $14.28\% = 1/7$**)

$$= (6/7) * 420 = 360$$

According to the question,

The number of children = 60

So, the sum of the number of males and females = $360 - 60 = 300$.

Let the number of males be 'c'

So, the number of females = $300 - c$

According to the question,

$$(30 * c) + 24 * (300 - c) = 8400$$

$$10c + 8 * (300 - c) = 2800$$

$$10c + 2400 - 8c = 2800$$

$$2c = 400$$

$$c = 200$$

The number of females in city U = $300 - c = 100$

4. Questions

The total number of males and females in,

$$\text{City P} = (100 * 2) = 200$$

$$\text{City Q} = (125 * 2) = 250$$

$$\text{City R} = (75 * 2) = 150$$

$$\text{City S} = (195 * 2) = 390$$

$$\text{City T} = (180 * 2) = 360$$

The total number of females and children in,

$$\text{City P} = (65 * 2) = 130$$

$$\text{City Q} = (75 * 2) = 150$$

$$\text{City R} = (45 * 2) = 90$$

$$\text{City S} = (90 * 2) = 180$$

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$$\text{City T} = (80 * 2) = 160$$

According to the note II,

Now,

Let the number of males in cities P and Q be $4k$ and $5k$ respectively.

So, the number of females in city P = $(200 - 4k)$

And, the number of children in city P = $\{130 - (200 - 4k)\} = (4k - 70)$

Also,

The number of females in city Q = $(250 - 5k)$

And, the number of children in city Q = $\{150 - (250 - 5k)\} = (5k - 100)$

Now,

$$(4k - 70) = (5k - 100)$$

$$k = 30$$

According to the note IV,

Also,

The ratio of the number of females in cities R and S = $40:100 = 2:5$. Let it be $2n$ and $5n$ respectively.

So, the number of males in city R = $(150 - 2n)$

And, the number of children in city R = $(90 - 2n)$

Also,

The number of males in city S = $(390 - 5n)$

And, the number of children in city S = $(180 - 5n)$

According to the note V,

Now,

$$(390 - 5n) - (150 - 2n) = 150$$

$$240 - 3n = 150$$

$$3n = 90$$

$$n = 30$$

City P

The number of males = $4k = (4 * 30) = 120$

The number of females = $(200 - 120) = 80$

The number of children = $(120 - 70) = 50$

The total number of people = $(120 + 80 + 50) = 250$

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City Q

The number of males = $5k = (5 * 30) = 150$

The number of females = $(250 - 150) = 100$

The number of children = $(150 - 100) = 50$

The total number of people = $(150 + 100 + 50) = 300$

City R

The number of females = $2n = (2 * 30) = 60$

The number of males = $(150 - 60) = 90$

The number of children = $(90 - 60) = 30$

The total number of people = $(60 + 90 + 30) = 180$

City S

The number of females = $5n = (5 * 30) = 150$

The number of males = $(390 - 150) = 240$

The number of children = $(180 - 150) = 30$

The total number of people = $(150 + 240 + 30) = 420$

City T

Let the number of females be m .

So, the number of males = $(360 - m)$

And, the number of children = $(160 - m)$

According to the note VI,

The total number of people = 400

Now,

$$m + (360 - m) + (160 - m) = 400$$

$$520 - m = 400$$

$$m = 120$$

Therefore,

The number of females = $m = 120$

The number of males = $(360 - m) = (360 - 120) = 240$

The number of children = $(160 - m) = (160 - 120) = 40$

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City	The total number of people	The number of males	The number of females	The number of children
P	250	120	80	50
Q	300	150	100	50
R	180	90	60	30
S	420	240	150	30
T	400	240	120	40
Total	1550	840	510	200

Answer: B

The total number of children in city R = 30

The total number of children in city T = 40

Let the number of children in secondary school in city R and T be x each.

So, the number of children in primary school in city R = 30 - x

And, the number of children in primary school in city T = 40 - x

Now,

$$(30 - x) / (40 - x) = 9/14$$

$$420 - 14x = 360 - 9x$$

$$5x = 60$$

$$x = 12$$

Therefore, the number of students in primary school in city R = (30 - x) = 30 - 12 = 18.

Hence, the correct answer is option B.

5. Questions

The total number of males and females in,

$$\text{City P} = (100 * 2) = 200$$

$$\text{City Q} = (125 * 2) = 250$$

$$\text{City R} = (75 * 2) = 150$$

$$\text{City S} = (195 * 2) = 390$$

$$\text{City T} = (180 * 2) = 360$$

The total number of females and children in,

$$\text{City P} = (65 * 2) = 130$$

$$\text{City Q} = (75 * 2) = 150$$

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$$\text{City R} = (45 * 2) = 90$$

$$\text{City S} = (90 * 2) = 180$$

$$\text{City T} = (80 * 2) = 160$$

According to the note II,

Now,

Let the number of males in cities P and Q be $4k$ and $5k$ respectively.

So, the number of females in city P = $(200 - 4k)$

And, the number of children in city P = $\{130 - (200 - 4k)\} = (4k - 70)$

Also,

The number of females in city Q = $(250 - 5k)$

And, the number of children in city Q = $\{150 - (250 - 5k)\} = (5k - 100)$

Now,

$$(4k - 70) = (5k - 100)$$

$$k = 30$$

According to the note IV,

Also,

The ratio of the number of females in cities R and S = $40:100 = 2:5$. Let it be $2n$ and $5n$ respectively.

So, the number of males in city R = $(150 - 2n)$

And, the number of children in city R = $(90 - 2n)$

Also,

The number of males in city S = $(390 - 5n)$

And, the number of children in city S = $(180 - 5n)$

According to the note V,

Now,

$$(390 - 5n) - (150 - 2n) = 150$$

$$240 - 3n = 150$$

$$3n = 90$$

$$n = 30$$

City P

The number of males = $4k = (4 * 30) = 120$

The number of females = $(200 - 120) = 80$

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The number of children = $(120 - 70) = 50$

The total number of people = $(120 + 80 + 50) = 250$

City Q

The number of males = $5k = (5 * 30) = 150$

The number of females = $(250 - 150) = 100$

The number of children = $(150 - 100) = 50$

The total number of people = $(150 + 100 + 50) = 300$

City R

The number of females = $2n = (2 * 30) = 60$

The number of males = $(150 - 60) = 90$

The number of children = $(90 - 60) = 30$

The total number of people = $(60 + 90 + 30) = 180$

City S

The number of females = $5n = (5 * 30) = 150$

The number of males = $(390 - 150) = 240$

The number of children = $(180 - 150) = 30$

The total number of people = $(150 + 240 + 30) = 420$

City T

Let the number of females be m .

So, the number of males = $(360 - m)$

And, the number of children = $(160 - m)$

According to the note VI,

The total number of people = 400

Now,

$$m + (360 - m) + (160 - m) = 400$$

$$520 - m = 400$$

$$m = 120$$

Therefore,

The number of females = $m = 120$

The number of males = $(360 - m) = (360 - 120) = 240$

The number of children = $(160 - m) = (160 - 120) = 40$

@QuizByMickey

City	The total number of people	The number of males	The number of females	The number of children
P	250	120	80	50
Q	300	150	100	50
R	180	90	60	30
S	420	240	150	30
T	400	240	120	40
Total	1550	840	510	200

Answer: A

In City T,

The total number of males and females = $240 + 120 = 360$

The total number of males and females who did not attend the seminar = $(1/4) * 360 = 90$

So, the total number of males and females who attended the seminar = $360 - 90 = 270$

The total number of males = 240

The number of males who attended the seminar = 200

So, the number of males who did not attend the seminar = $240 - 200 = 40$

The total number of females = 120

The number of females who attended the seminar = $270 - 200 = 70$

So, the number of females who did not attend the seminar = $120 - 70 = 50$

Therefore, the required percentage = $(50/120) * 100 = 41.67\%$.

Hence, the correct answer is option A.

6. Questions

Answer: B

Using I,

The total investment made by A = $(1500 * 12) = \text{Rs. } 18000$

The total investment made by B = $(140/100) * 1500 * 5 = 2100 * 5 = \text{Rs. } 10500$

The ratio of the profit share received by A and B = $18000:10500 = 12:7$.

So, statement I is satisfied.

Using II

The total investment made by A = $(1500 * 12) = \text{Rs. } 18000$

The total investment made by B = $(120/100) * 1500 * 4 = 1800 * 4 = \text{Rs. } 7200$

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The ratio of the profit share received by A and B = $18000:7200 = 5:2$.

So, statement II is satisfied.

Using III

The total investment made by A = $(1500 * 12) = \text{Rs. } 18000$

The total investment made by B = $(180/100) * 1500 * 6 = (2700 * 6) = \text{Rs. } 16200$

The ratio of the profit share received by A and B = $18000:16200 = 10:9$.

So, statement III does not satisfy.

Therefore, both I and II follow.

Hence, the correct answer is option B.

7. Questions

Answer: A

Using I

The Cost Price of the article = Rs. 2000

The Selling price of the article = $(135/100) * 2000 = \text{Rs. } 2700$

The Marked price of the article = $(2700 + 300) = \text{Rs. } 3000$

So, statement I is satisfied.

Using II

The Cost Price of the article = Rs. 1000

The Selling price of the article = $(180/100) * 1000 = \text{Rs. } 1800$

The Marked price of the article = $(1800 + 300) = \text{Rs. } 2100$

So, statement II is not satisfied.

Using III

The Cost Price of the article = Rs. 4000

The Selling price of the article = $(120/100) * 4000 = \text{Rs. } 4800$

The Marked price of the article = $(4800 + 300) = \text{Rs. } 5100$

So, statement III is not satisfied.

Therefore, only I follow.

Hence, the correct answer is option A.

8. Questions

Answer: C

Using I

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The speed of the river = 6 km/hr

The speed of the boat in still water = $(6 * 4) = 24$ km/hr

The downstream speed of the boat = $(24 + 6) = 30$ km/hr

The upstream speed of the boat = $(24 - 6) = 18$ km/hr

Now,

The time taken by the boat to cover downstream distance = $(120/30) = 4$ hours

The time taken by the boat to cover upstream distance = $(60/18) = 3.33$ hours

So, statement I is not satisfied.

Using II

The speed of the river = 8 km/hr

The speed of the boat in still water = $(8 * 4) = 32$ km/hr

The downstream speed of the boat = $(32 + 8) = 40$ km/hr

The upstream speed of the boat = $(32 - 8) = 24$ km/hr

Now,

The time taken by the boat to cover downstream distance = $(120/40) = 3$ hours

The time taken by the boat to cover upstream distance = $(72/24) = 3$ hours

So, statement II is satisfied.

Using III

The speed of the river = 4 km/hr

The speed of the boat in still water = $(4 * 4) = 16$ km/hr

The downstream speed of the boat = $(16 + 4) = 20$ km/hr

The upstream speed of the boat = $(16 - 4) = 12$ km/hr

Now,

The time taken by the boat to cover downstream distance = $(120/20) = 6$ hours

The time taken by the boat to cover upstream distance = $(48/12) = 4$ hours

So, statement III is not satisfied.

Therefore, only II follow.

Hence, the correct answer is option C.

9. Questions

Answer: B

The amount invested in scheme A = Rs. 40000

Using I

The simple interest received from scheme A = $(12/100) * 4 * 40000 = (48/100) * 40000 = \text{Rs. } 19200$

The amount invested in scheme B = Rs. 19200

The compound interest received from scheme B = $19200 * [(120/100) * (120/100) - 1] = 19200 * (44/100) = \text{Rs. } 8448$.

So, statement I is satisfied.

Using II

The simple interest received from scheme A = $(16/100) * 4 * 40000 = 64 * 400 = \text{Rs. } 25600$

The amount invested in scheme B = Rs. 25600

The compound interest received from scheme B = $25600 * [(120/100) * (120/100) - 1] = (44/100) \text{ of } 25600 = \text{Rs. } 11264$

So, statement II is not satisfied.

Using III

The simple interest received from scheme A = $(8/100) * 4 * 40000 = 32 * 400 = \text{Rs. } 12800$

The amount invested in scheme B = Rs. 12800

The compound interest received from scheme B = $12800 * [(120/100) * (120/100) - 1] = (44/100) * 12800 = \text{Rs. } 5632$

So, statement III is satisfied.

Therefore, both I and III follow.

Hence, the correct answer is option B.

10. Questions

Answer: D

Using I

The length of train P = 240 meters

The length of train Q = 300 meters

So, the time taken by train P to cross the platform = $\{(240 + 120)/40\} = (360/40) = 9 \text{ seconds}$

And, the time taken by train Q to cross the platform = $\{(300 + 120)/60\} = (420/60) = 7 \text{ seconds}$

So, the required difference = $(9 - 7) = 2 \text{ seconds } (P > Q)$

Therefore, statement I is satisfied.

Using II

The length of train P = 280 meters

The length of train Q = 360 meters

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So, the time taken by train P to cross the platform = $\{(280 + 120)/40\} = (400/40) = 10$ seconds

And, the time taken by train Q to cross the platform = $\{(360+120)/60\} = (480/60) = 8$ seconds

So, the required difference = $(10 - 8) = 2$ seconds ($P > Q$)

Therefore, statement II is satisfied.

Using III

The length of train P = 360 meters

The length of train Q = 480 meters

So, the time taken by train P to cross the platform = $\{(360 + 120)/40\} = (480/40) = 12$ seconds

And, the time taken by train Q to cross the platform = $\{(480 + 120)/60\} = (600/60) = 10$ seconds

The required difference = $(12 - 10) = 2$ seconds ($P > Q$)

So, statement III is satisfied.

Therefore, All I, II and III follow

Hence, the correct answer is option D.

11. Questions

Let the total number of cars sold by all five shops together be 500k.

So, the total number of diesel cars sold by all five shops together = $(2/5) * 500k = 200k$

Now,

The total number of cars sold by shop A = $(24/100) * 500k = 120k$

The number of diesel cars sold by shop A = $(28/100) * 200k = 56k$

The sum of the number of petrol and electric cars sold by shop A = $(120k - 56k) = 64k$

The number of electric cars sold by shop A = 144

The number of petrol cars sold by shop A = $(64k - 144)$

Also,

The total number of cars sold by shop B = $(32/100) * 500k = 160k$

The number of diesel cars sold by shop B = $(30/100) * 200k = 60k$

The sum of the number of petrol and electric cars sold by shop B = $(160k - 60k) = 100k$

The number of electric cars sold by shop B = 200

The number of petrol cars sold by shop B = $(100k - 200)$

So,

$$(100k - 200) - (64k - 144) = 160$$

$$100k - 360 = 64k - 144$$

$$36k = 216$$

$$k = 6$$

So, the total number of cars sold by all five shops together = $500k = (500 * 6) = 3000$

And, the total number of diesel cars sold by all five shops together = $200k = (200 * 6) = 1200$

In Shop A,

The total number of cars sold = $(24/100) * 3000 = 720$

The number of diesel cars sold = $(28/100) * 1200 = 336$

The sum of the number of petrol and electric cars sold = $(720 - 336) = 384$

The number of electric cars sold = 144

The number of petrol cars sold = $(384 - 144) = 240$

In Shop B,

The total number of cars sold = $(32/100) * 3000 = 960$

The number of diesel cars sold = $(30/100) * 1200 = 360$

The sum of the number of petrol and electric cars sold = $(960 - 360) = 600$

The number of electric cars sold = 200

The number of petrol cars sold = $(600 - 200) = 400$

In Shop C,

The total number of cars sold = $(16/100) * 3000 = 480$

The number of diesel cars sold = $(15/100) * 1200 = 180$

The sum of the number of petrol and electric cars sold = $(480 - 180) = 300$

The number of electric cars sold = 60

The number of petrol cars sold = $(300 - 60) = 240$

In Shop D,

The total number of cars sold = $(20/100) * 3000 = 600$

The number of diesel cars sold = $(19/100) * 1200 = 228$

The sum of the number of petrol and electric cars sold = $(600 - 228) = 372$

The number of electric cars sold = 124

The number of petrol cars sold = $(372 - 124) = 248$

In Shop E,

The total number of cars sold = $(8/100) * 3000 = 240$

The number of diesel cars sold = $(8/100) * 1200 = 96$

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The sum of the number of petrol and electric cars sold = $(240 - 96) = 144$

The number of electric cars sold = 24

The number of petrol cars sold = $(144 - 24) = 120$

Shop	The total number of cars sold	The number of petrol cars sold	The number of diesel cars sold	The number of electric cars sold
A	720	240	336	144
B	960	400	360	200
C	480	240	180	60
D	600	248	228	124
E	240	120	96	24
Total	3000	1248	1200	552

Answer: A

Now, according to the question,

The total number of petrol cars sold by shop E = 120

The number of petrol cars sold to females by shop E = $(25/100) * 120 = 30$

And, the number of petrol cars sold to males by shop E = $(120 - 30) = 90$

So, the number of petrol cars sold to females by shop C = $\{(30/3) * 8\} = 80$

The total number of petrol cars sold by shop C = 240

So, the number of petrol cars sold to males by shop C = $(240 - 80) = 160$

Therefore, the required difference = $(160 - 90) = 70$.

Hence, the correct answer is option A.

12. Questions

Let the total number of cars sold by all five shops together be 500k.

So, the total number of diesel cars sold by all five shops together = $(2/5) * 500k = 200k$

Now,

The total number of cars sold by shop A = $(24/100) * 500k = 120k$

The number of diesel cars sold by shop A = $(28/100) * 200k = 56k$

The sum of the number of petrol and electric cars sold by shop A = $(120k - 56k) = 64k$

The number of electric cars sold by shop A = 144

The number of petrol cars sold by shop A = $(64k - 144)$

Also,

The total number of cars sold by shop B = $(32/100) * 500k = 160k$

The number of diesel cars sold by shop B = $(30/100) * 200k = 60k$

The sum of the number of petrol and electric cars sold by shop B = $(160k - 60k) = 100k$

The number of electric cars sold by shop B = 200

The number of petrol cars sold by shop B = $(100k - 200)$

So,

$$(100k - 200) - (64k - 144) = 160$$

$$100k - 360 = 64k - 144$$

$$36k = 216$$

$$k = 6$$

So, the total number of cars sold by all five shops together = $500k = (500 * 6) = 3000$

And, the total number of diesel cars sold by all five shops together = $200k = (200 * 6) = 1200$

In Shop A,

The total number of cars sold = $(24/100) * 3000 = 720$

The number of diesel cars sold = $(28/100) * 1200 = 336$

The sum of the number of petrol and electric cars sold = $(720 - 336) = 384$

The number of electric cars sold = 144

The number of petrol cars sold = $(384 - 144) = 240$

In Shop B,

The total number of cars sold = $(32/100) * 3000 = 960$

The number of diesel cars sold = $(30/100) * 1200 = 360$

The sum of the number of petrol and electric cars sold = $(960 - 360) = 600$

The number of electric cars sold = 200

The number of petrol cars sold = $(600 - 200) = 400$

In Shop C,

The total number of cars sold = $(16/100) * 3000 = 480$

The number of diesel cars sold = $(15/100) * 1200 = 180$

The sum of the number of petrol and electric cars sold = $(480 - 180) = 300$

The number of electric cars sold = 60

The number of petrol cars sold = $(300 - 60) = 240$

In Shop D,

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The total number of cars sold = $(20/100) * 3000 = 600$

The number of diesel cars sold = $(19/100) * 1200 = 228$

The sum of the number of petrol and electric cars sold = $(600 - 228) = 372$

The number of electric cars sold = 124

The number of petrol cars sold = $(372 - 124) = 248$

In Shop E,

The total number of cars sold = $(8/100) * 3000 = 240$

The number of diesel cars sold = $(8/100) * 1200 = 96$

The sum of the number of petrol and electric cars sold = $(240 - 96) = 144$

The number of electric cars sold = 24

The number of petrol cars sold = $(144 - 24) = 120$

Shop	The total number of cars sold	The number of petrol cars sold	The number of diesel cars sold	The number of electric cars sold
A	720	240	336	144
B	960	400	360	200
C	480	240	180	60
D	600	248	228	124
E	240	120	96	24
Total	3000	1248	1200	552

Answer: D

In Shop B,

Let the total number of petrol cars manufactured be N.

So, the total number of diesel cars manufactured = $(N - 20)$

The number of petrol cars sold = 400

The number of diesel cars sold = 360

Now,

$$(N - 400) / (N - 20 - 360) = 5/6$$

$$(N - 400) / (N - 380) = 5/6$$

$$6N - 2400 = 5N - 1900$$

$$N = 500$$

Therefore, the total number of diesel cars manufactured by shop B = $(N - 20) = (500 - 20) = 480$.

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Hence, the correct answer is option D.

13. Questions

Let the total number of cars sold by all five shops together be 500k.

So, the total number of diesel cars sold by all five shops together = $(2/5) * 500k = 200k$

Now,

The total number of cars sold by shop A = $(24/100) * 500k = 120k$

The number of diesel cars sold by shop A = $(28/100) * 200k = 56k$

The sum of the number of petrol and electric cars sold by shop A = $(120k - 56k) = 64k$

The number of electric cars sold by shop A = 144

The number of petrol cars sold by shop A = $(64k - 144)$

Also,

The total number of cars sold by shop B = $(32/100) * 500k = 160k$

The number of diesel cars sold by shop B = $(30/100) * 200k = 60k$

The sum of the number of petrol and electric cars sold by shop B = $(160k - 60k) = 100k$

The number of electric cars sold by shop B = 200

The number of petrol cars sold by shop B = $(100k - 200)$

So,

$$(100k - 200) - (64k - 144) = 160$$

$$100k - 360 = 64k - 144$$

$$36k = 216$$

$$k = 6$$

So, the total number of cars sold by all five shops together = $500k = (500 * 6) = 3000$

And, the total number of diesel cars sold by all five shops together = $200k = (200 * 6) = 1200$

In Shop A,

The total number of cars sold = $(24/100) * 3000 = 720$

The number of diesel cars sold = $(28/100) * 1200 = 336$

The sum of the number of petrol and electric cars sold = $(720 - 336) = 384$

The number of electric cars sold = 144

The number of petrol cars sold = $(384 - 144) = 240$

In Shop B,

The total number of cars sold = $(32/100) * 3000 = 960$

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The number of diesel cars sold = $(30/100) * 1200 = 360$

The sum of the number of petrol and electric cars sold = $(960 - 360) = 600$

The number of electric cars sold = 200

The number of petrol cars sold = $(600 - 200) = 400$

In Shop C,

The total number of cars sold = $(16/100) * 3000 = 480$

The number of diesel cars sold = $(15/100) * 1200 = 180$

The sum of the number of petrol and electric cars sold = $(480 - 180) = 300$

The number of electric cars sold = 60

The number of petrol cars sold = $(300 - 60) = 240$

In Shop D,

The total number of cars sold = $(20/100) * 3000 = 600$

The number of diesel cars sold = $(19/100) * 1200 = 228$

The sum of the number of petrol and electric cars sold = $(600 - 228) = 372$

The number of electric cars sold = 124

The number of petrol cars sold = $(372 - 124) = 248$

In Shop E,

The total number of cars sold = $(8/100) * 3000 = 240$

The number of diesel cars sold = $(8/100) * 1200 = 96$

The sum of the number of petrol and electric cars sold = $(240 - 96) = 144$

The number of electric cars sold = 24

The number of petrol cars sold = $(144 - 24) = 120$

Shop	The total number of cars sold	The number of petrol cars sold	The number of diesel cars sold	The number of electric cars sold
A	720	240	336	144
B	960	400	360	200
C	480	240	180	60
D	600	248	228	124
E	240	120	96	24
Total	3000	1248	1200	552

Answer: B

In Shop F,

The total number of cars sold = $(600 - 100) = 500$

The number of petrol cars sold = 250

The sum of the number of diesel and electric cars sold = $500 - 250 = 250$

The sum of the number of petrol and diesel cars sold = $250 * (8/5) = 400$

The number of diesel cars sold = $400 - 250 = 150$

The number of electric cars sold = $250 - 150 = 100$

Hence, the correct answer is option B.

14. Questions

Let the total number of cars sold by all five shops together be 500k.

So, the total number of diesel cars sold by all five shops together = $(2/5) * 500k = 200k$

Now,

The total number of cars sold by shop A = $(24/100) * 500k = 120k$

The number of diesel cars sold by shop A = $(28/100) * 200k = 56k$

The sum of the number of petrol and electric cars sold by shop A = $(120k - 56k) = 64k$

The number of electric cars sold by shop A = 144

The number of petrol cars sold by shop A = $(64k - 144)$

Also,

The total number of cars sold by shop B = $(32/100) * 500k = 160k$

The number of diesel cars sold by shop B = $(30/100) * 200k = 60k$

The sum of the number of petrol and electric cars sold by shop B = $(160k - 60k) = 100k$

The number of electric cars sold by shop B = 200

The number of petrol cars sold by shop B = $(100k - 200)$

So,

$$(100k - 200) - (64k - 144) = 160$$

$$100k - 360 = 64k - 144$$

$$36k = 216$$

$$k = 6$$

So, the total number of cars sold by all five shops together = $500k = (500 * 6) = 3000$

And, the total number of diesel cars sold by all five shops together = $200k = (200 * 6) = 1200$

In Shop A,

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The total number of cars sold = $(24/100) * 3000 = 720$

The number of diesel cars sold = $(28/100) * 1200 = 336$

The sum of the number of petrol and electric cars sold = $(720 - 336) = 384$

The number of electric cars sold = 144

The number of petrol cars sold = $(384 - 144) = 240$

In Shop B,

The total number of cars sold = $(32/100) * 3000 = 960$

The number of diesel cars sold = $(30/100) * 1200 = 360$

The sum of the number of petrol and electric cars sold = $(960 - 360) = 600$

The number of electric cars sold = 200

The number of petrol cars sold = $(600 - 200) = 400$

In Shop C,

The total number of cars sold = $(16/100) * 3000 = 480$

The number of diesel cars sold = $(15/100) * 1200 = 180$

The sum of the number of petrol and electric cars sold = $(480 - 180) = 300$

The number of electric cars sold = 60

The number of petrol cars sold = $(300 - 60) = 240$

In Shop D,

The total number of cars sold = $(20/100) * 3000 = 600$

The number of diesel cars sold = $(19/100) * 1200 = 228$

The sum of the number of petrol and electric cars sold = $(600 - 228) = 372$

The number of electric cars sold = 124

The number of petrol cars sold = $(372 - 124) = 248$

In Shop E,

The total number of cars sold = $(8/100) * 3000 = 240$

The number of diesel cars sold = $(8/100) * 1200 = 96$

The sum of the number of petrol and electric cars sold = $(240 - 96) = 144$

The number of electric cars sold = 24

The number of petrol cars sold = $(144 - 24) = 120$

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Shop	The total number of cars sold	The number of petrol cars sold	The number of diesel cars sold	The number of electric cars sold
A	720	240	336	144
B	960	400	360	200
C	480	240	180	60
D	600	248	228	124
E	240	120	96	24
Total	3000	1248	1200	552

Answer: C

In Shop E,

The total number of cars sold = 240

So, the number of cars unsold = $(240 - 80) = 160$

The total number of cars manufactured = $(240 + 160) = 400$

Therefore, the required percentage = $(160/400) * 100 = 40\%$

So, ? = 40.

Hence, the correct answer is option C.

15. Questions

Let the total number of cars sold by all five shops together be 500k.

So, the total number of diesel cars sold by all five shops together = $(2/5) * 500k = 200k$

Now,

The total number of cars sold by shop A = $(24/100) * 500k = 120k$

The number of diesel cars sold by shop A = $(28/100) * 200k = 56k$

The sum of the number of petrol and electric cars sold by shop A = $(120k - 56k) = 64k$

The number of electric cars sold by shop A = 144

The number of petrol cars sold by shop A = $(64k - 144)$

Also,

The total number of cars sold by shop B = $(32/100) * 500k = 160k$

The number of diesel cars sold by shop B = $(30/100) * 200k = 60k$

The sum of the number of petrol and electric cars sold by shop B = $(160k - 60k) = 100k$

The number of electric cars sold by shop B = 200

The number of petrol cars sold by shop B = $(100k - 200)$

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So,

$$(100k - 200) - (64k - 144) = 160$$

$$100k - 360 = 64k - 144$$

$$36k = 216$$

$$k = 6$$

So, the total number of cars sold by all five shops together = $500k = (500 * 6) = 3000$

And, the total number of diesel cars sold by all five shops together = $200k = (200 * 6) = 1200$

In Shop A,

The total number of cars sold = $(24/100) * 3000 = 720$

The number of diesel cars sold = $(28/100) * 1200 = 336$

The sum of the number of petrol and electric cars sold = $(720 - 336) = 384$

The number of electric cars sold = 144

The number of petrol cars sold = $(384 - 144) = 240$

In Shop B,

The total number of cars sold = $(32/100) * 3000 = 960$

The number of diesel cars sold = $(30/100) * 1200 = 360$

The sum of the number of petrol and electric cars sold = $(960 - 360) = 600$

The number of electric cars sold = 200

The number of petrol cars sold = $(600 - 200) = 400$

In Shop C,

The total number of cars sold = $(16/100) * 3000 = 480$

The number of diesel cars sold = $(15/100) * 1200 = 180$

The sum of the number of petrol and electric cars sold = $(480 - 180) = 300$

The number of electric cars sold = 60

The number of petrol cars sold = $(300 - 60) = 240$

In Shop D,

The total number of cars sold = $(20/100) * 3000 = 600$

The number of diesel cars sold = $(19/100) * 1200 = 228$

The sum of the number of petrol and electric cars sold = $(600 - 228) = 372$

The number of electric cars sold = 124

The number of petrol cars sold = $(372 - 124) = 248$

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In Shop E,

The total number of cars sold = $(8/100) * 3000 = 240$

The number of diesel cars sold = $(8/100) * 1200 = 96$

The sum of the number of petrol and electric cars sold = $(240 - 96) = 144$

The number of electric cars sold = 24

The number of petrol cars sold = $(144 - 24) = 120$

Shop	The total number of cars sold	The number of petrol cars sold	The number of diesel cars sold	The number of electric cars sold
A	720	240	336	144
B	960	400	360	200
C	480	240	180	60
D	600	248	228	124
E	240	120	96	24
Total	3000	1248	1200	552

Answer: B

In Shop A,

The ratio of the number of defective and non-defective diesel cars manufactured = $1: (5 - 1) = 1:4$.

The number of diesel cars sold = 336 (all non-defective)

The number of non-defective diesel cars unsold = 144

So, the total number of non-defective cars (sold + unsold) = $(336 + 144) = 480$

Therefore, the total number of diesel cars manufactured by shop A = $480 * (5/4) = 600$.

Hence, the correct answer is option B.

16. Questions

Answer: B

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$(29.99\% \text{ of } 720.01) + (40.03\% \text{ of } 359.97) = ? + (49.89\% \text{ of } 499.92) + (10.08\% \text{ of } 1000.06)$

$(30\% \text{ of } 720) + (40\% \text{ of } 360) = ? + (50\% \text{ of } 500) + (10\% \text{ of } 1000)$

$216 + 144 = ? + 250 + 100$

$360 = ? + 350$

$? = 10$

Hence, the correct answer is option B.

17. Questions

Answer: A

$$(23.95 * 8.06) \div (12.09 * 3.93) + ? = (34.98 * 4.96) - (23.04 * 7.01)$$

$$(24 * 8) \div (12 * 4) + ? = (35 * 5) - (23 * 7)$$

$$192 \div 48 + ? = 175 - 161$$

$$4 + ? = 14$$

$$? = 10$$

Hence, the correct answer is option A.

18. Questions

Answer: C

$$(\sqrt{24.89\% \text{ of } 1200.11}) + (\sqrt{16.13\% \text{ of } 999.88}) = (12.09\% \text{ of } 499.96) + (19.91\% \text{ of } 400.09) - ?$$

$$(\sqrt{25\% \text{ of } 1200}) + (\sqrt{16\% \text{ of } 1000}) = (12\% \text{ of } 500) + (20\% \text{ of } 400) - ?$$

$$(5\% \text{ of } 1200) + (4\% \text{ of } 1000) = 60 + 80 - ?$$

$$60 + 40 = 140 - ?$$

$$100 = 140 - ?$$

$$? = 40$$

Hence, the correct answer is option C.

19. Questions

Answer: D

$$(63.91/15.94) * (90.04/9.02) = ? + [(87.93/3.98) * (45.01/9.07)]$$

$$(64/16) * (90/9) = ? + [(88/4) * (45/9)]$$

$$4 * 10 = ? + (22 * 5)$$

$$40 = ? + 110$$

$$40 - 110 = ?$$

$$? = -70$$

Hence, the correct answer is option D.

20. Questions

Answer: E

$$[(390.01/2.99) + (119.93/4.07)] - [(249.96/5.11) + (140.06/13.90)] = ?^2$$

$$[(390/3) + (120/4)] - [(250/5) + (140/14)] = ?^2$$

$$(130 + 30) - (50 + 10) = ?^2$$

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$$160 - 60 = ?^2$$

$$100 = ?^2$$

$$\sqrt{100} = ?$$

$$10 = ?$$

Hence, the correct answer is option E.

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